Cost of Inaction

Cybersecurity Risk & Economic Impact Analysis

Prepared April 24, 2026

Executive document — for internal use only.

1. Executive Summary

This report assesses the economic cost of maintaining the current cybersecurity posture across 2 active risks. It compares three investment scenarios — no action, partial investment, and strategic investment — and quantifies the financial difference between each path.

Cost of inaction (expected loss): 0.15K | Worst-case stress scenario (P90): 0.19K

1 risk is currently classified as HIGH or CRITICAL. These risks are the primary drivers of expected loss and the first candidates for additional investment.

Best strategy: No Investment — avoids 0.0K in expected losses at a cost of 0.0K, delivering a net benefit of 0.0K and a 0.0% reduction in portfolio loss.

Cybersecurity investment should be understood as a value-preserving decision: the expected return exceeds the cost, and the alternative is accepting avoidable losses. The sections below provide the full evidence behind this conclusion.

2. Risk Overview

The assessment covers 2 active risks evaluated on their residual exposure — the remaining risk after applying existing controls. Residual risk is the baseline for all economic estimates in this report.

Distribution by level: 1 MEDIUM, 1 HIGH

Residual risk scores and individual economic loss contribution, ordered from highest to lowest impact:

- Human errors: residual score 11.00 (HIGH) — expected loss 0.113K
- Service interruption: residual score 5.70 (MEDIUM) — expected loss 0.016K

The 1 HIGH/CRITICAL risk listed above account for a disproportionate share of portfolio loss. Current controls reduce their exposure, but residual levels remain above acceptable tolerance — indicating a gap between existing mitigation and what is needed to reach a defensible posture.

3. Scenario Analysis

This section compares three investment scenarios on expected loss, statistical outcome ranges, and total cost. The P50 figure is the median simulated outcome; P90 is the threshold exceeded only in the worst 10% of scenarios — the relevant benchmark for stress-testing organizational resilience and financial planning.

Total expected cost (loss plus investment) is the most complete comparison basis. A scenario with higher upfront investment but lower total cost is financially superior when the avoided loss exceeds the additional spending.

How to read these figures

The risk scores and rankings in the previous section are calculated directly from the assessment model — each number is a fixed value derived from probability, impact, and control effectiveness. The figures below (expected loss, P50, P90) come from a different source: thousands of randomized what-if scenarios run against the same risk profile. They describe a range of plausible outcomes, not a single forecast. A higher P90 does not mean the risk scores were wrong — it means adverse outcomes, though unlikely, carry a larger financial consequence than the average suggests.

No Investment

Current posture maintained. No additional controls or mitigation programs are applied. Exposure reflects the unmodified residual risk profile.

- Expected loss (mean): 0.15K
- Median outcome (P50): 0.15K
- Stress scenario (P90): 0.19K — tail spread vs P50: 0.04K
- Investment required: 0.0K
- Total expected cost (loss + investment): 0.15K

Partial Investment

Targeted improvements applied to the highest-risk areas. Reduces event likelihood on priority risks, but leaves structural gaps in place across the broader portfolio.

- Expected loss (mean): 0.06K
- Median outcome (P50): 0.06K
- Stress scenario (P90): 0.08K — tail spread vs P50: 0.02K
- Investment required: 0.4K
- Total expected cost (loss + investment): 0.46K

Strategic Investment

Comprehensive program aligned with the full risk profile. Addresses both event likelihood and potential impact systematically across all identified risks.

- Expected loss (mean): 0.01K
- Median outcome (P50): 0.01K
- Stress scenario (P90): 0.02K — tail spread vs P50: 0.01K
- Investment required: 1.0K
- Total expected cost (loss + investment): 1.01K

A wide P50-to-P90 spread signals high tail risk: while the typical outcome may be manageable, adverse scenarios carry disproportionately higher losses. Organizations with low risk tolerance should weight P90 more heavily when selecting a scenario.

Lowest total expected cost: No Investment

4. Economic Impact

This section quantifies the total economic exposure of the portfolio and identifies which risks drive the largest share of expected losses. These figures directly inform prioritization decisions: where to invest first for maximum loss reduction.

Portfolio expected loss (current posture): 0.15K

- Median simulated outcome (P50): 0.15K
- Stress scenario (P90): 0.19K — the organization should be financially prepared to absorb this amount

Top risks by economic contribution

The three risks with the highest individual expected loss:

1. Human errors — expected loss 0.113K (75.3% of total) [HIGH]
2. Service interruption — expected loss 0.016K (10.7% of total) [MEDIUM]

These three risks account for approximately 86% of total portfolio loss. Targeted investment here delivers the highest loss-reduction return per unit of spending.

Avoided loss under No Investment: 0.0K | Net benefit after investment: 0.0K

All monetary figures use a normalized K-unit scale that enables scenario comparison without reference to absolute financial values. The proportional relationships between scenarios remain accurate regardless of the K denomination used.

5. Recommendations

Primary recommendation: implement the No Investment scenario.

This strategy produces a net financial benefit of 0.0K — the expected losses avoided (0.0K) exceed the investment required (0.0K). It reduces expected portfolio loss by 0.0% relative to the current posture.

Prioritization: where to invest first

Based on economic impact analysis, investment should be concentrated on the following risks in order of priority:

1. Human errors — expected loss 0.113K [HIGH]
2. Service interruption — expected loss 0.016K [MEDIUM]

Improving controls on these risks delivers the greatest reduction in total portfolio loss. Remaining risks should be addressed in a subsequent phase once primary mitigations are confirmed effective.

Planning assumptions and limitations

These estimates are based on a Monte Carlo simulation with controlled uncertainty. Results represent the likely range of outcomes — they are not accounting forecasts. Material changes to the risk environment (new threats, regulatory changes, organizational restructuring) would warrant a revised assessment.

The P90 exposure of 0.19K under the current posture should be treated as a financial planning ceiling — a conservative buffer against adverse scenarios. Organizations with low risk tolerance should ensure this amount is available as a contingency reserve.

Investment decisions should be revisited on a regular cycle (at minimum annually) to reflect changes in the risk profile, control effectiveness, and strategic priorities.